

PROJECT 3: Ames Housing - Predicting Real Estate Prices

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▼ Deliverables

- An `.Rmd` file.
- The corresponding knitted HTML file.
- The presentation slides (submitted as a PDF).

▼ Instructions

For the Rmd/HTML report, explanations must be provided for each step. The layout must be professional and "business-ready," paying close attention to Markdown formatting, the use of a CSS file for styling, and the inclusion of a Table of Contents (TOC).

Regarding the presentation, the total speaking time is 10 minutes (to be shared), followed by a 5-minute Q&A session. The presentation support must be slides in PPTX format; while there is no maximum slide count, there should be at least 6 slides at the minimum. The slides must be submitted in PDF format.

It would be appreciated if you use a Notion/Git or any project tool ; that will give you bonus points.

▼ Recommendations

Pay special attention to the polish and formatting.. The presentation should concentrate on a quick overview of the case study, its analyses (focusing on

the **results**, not the technical operations), its visualizations, and its conclusions, including different avenues for planned or future improvements.

▼ Timeline

- Finalisation of the groups : Saturday 18/10 23H59
- Start of the project : Sunday 19/10 00H00
- Next session when you can ask any questions : Wednesday 22/10
- Time limit of the submission of project files (Rmd/HTML) : Monday 10/11 at 23h00
- Time limit of the submission of presentation file (pdf) : Wednesday 12/11 at 7h00
- Presentation of the project : Wednesday 12/11

For each student , you'll have to submit the files : just submit one version of it on Junia Learning ; then for each group, you'll tell me which member has the latest version of the project.

Project Goal : As analysts for a real estate agency, your team must develop a model to predict the sale price of homes in Ames, Iowa. This dataset is rich but complex, with many features and missing values that have different meanings. Your challenge will be to untangle this complexity to build an accurate regression model.

Source : <https://www.kaggle.com/c/house-prices-advanced-regression-techniques/data>

Step 1: Data Cleaning and Preparation

- By looking at the meta-data associated to this dataset, look how to handle the NA values by the nature of column and apply the proper operations
 - Be careful ; some columns don't have the same signification for NA values.
- Identify columns where the vast majority of values are the same (e.g., **Street** or **Utilities**). Discuss their relevance and consider removing them.

- **Outlier Detection :** The dataset documentation mentions outliers. Create a scatter plot of `GrLivArea` (above-ground living area) vs. `SalePrice` and identify the few points that deviate heavily from the trend to remove them.

Step 2: Feature Transformation and Engineering

- **Logarithmic Transformation:** The distribution of `SalePrice` is highly skewed. Apply a log transformation to this variable to normalize it, which often improves the performance of regression models.
- **Creating Surface Area Features:** Create a `TotalSF` feature which corresponds to the total house area.
- **Encoding Ordinal Variables:** Many categorical features are actually ordinal. Convert them to numerical variables so the model can understand their inherent order.
- **Creating Temporal Features:** Create a `HouseAge` variable by calculating the difference between the year sold and the year built.

Step 3: Visualization and Exploration (with `ggplot2`)

- **Correlation with Sale Price:** Calculate the correlation of all numerical variables with `SalePrice` . Create a bar chart to visualize the top 15 most correlated features.
- **Impact of Overall Quality:** Use box plots to show the relationship between `OverallQual` and `SalePrice` .
- **Price by Neighborhood:** Create box plots to compare the distribution of `SalePrice` across the different `Neighborhood` .
- **Size-Price Relationship:** Make a scatter plot (`geom_point`) of `SalePrice` vs. your new `TotalSF` feature, coloring the points by `OverallQual` .
- Add 2 pertinent visualisations of your liking.

Step 4: Modeling and Prediction (Data Science)

- **Multiple Linear Regression:** Train a linear regression model as a baseline, using a selection of your best features.
- **Regularized Regression (LASSO):** Train a LASSO regression model (with the `glmnet` package). The advantage of LASSO is that it can perform automatic feature selection by shrinking the coefficients of less important variables to zero.
- **Tree-Based Model:** Train a Random Forest model, which is often very effective for this type of problem.
- **Model Evaluation:** Evaluate the performance of your models using multiple indicators on the predictions. Remember to transform your predictions back from the log scale before calculating the error. Interpret the most important features from your best model.